

Artifact 23 — Agentic Logic Specification

Execution · Stage B · The Engineering Blueprint — Logic

DOCUMENT DETAILS

Project	Home-Care-AI	Stage	Execution → Stage B (The Engineering Blueprint — Logic)
Date	2026-03-28	Author Role	Technical PM & System Architect
Feature	Vacancy Incident Processing Pipeline — Smart Match + HITL Approval + Notification Dispatch		
Regulatory	Australian Privacy Act 1988 (APP). HIPAA-grade security as design floor. All data in AWS ap-southeast-2 (Sydney). No cross-border transmission in v1.		

Handshake Inputs Consumed

ID	Source	What Transferred
HS-STRAT-01	Artifact 15 §4 (Moment of Truth) + §6 (L3 Interventions)	All L3 actions (ACT-C-01, ACT-C-02, ACT-P-01, ACT-F-01, VACANCY_UNRESOLVED) → Gates 8–14
HS-STRAT-02a-i	Artifact 16 §9 (Mitigation Requirements)	NFR-01–06 + compound guards CC-1, CC-4, CC-6, CC-8, G-E3-1, G-DS-05
HS-STRAT-03	Artifact 20 §12 (Token Budget)	Haiku → L1/L2 (≤2,200 tokens); Sonnet → L3 only (≤3,500 tokens); No LLM → notifications
HS-STRAT-04	Artifact 14 §6 (OKRs)	OKR-1 through OKR-6 → success criteria for each gate
Artifact 22 BS-1	Artifact 16 §7 canonical event registry	31-event registry is canonical — this spec uses those event names only
Artifact 22 BS-2	F10 dormant schema fields	credential_expiry_date, care_plan_review_due_date, visit.documentation_complete planted in Wave 1 schema
Artifact 22 BS-4	VACANCY_UNRESOLVED threshold	VACANCY_UNRESOLVED_ALERT_THRESHOLD = 0.10 (10%/14 days) → PM Lead L1 alert
Artifact 22 BS-5	EC-04 cold-start definition	SPP cold-start (P-10 = 0) is EC-04 — distinct from EC-02 (roster exhaustion)

Feeds into

- user-stories (Execution Stage B, Skill 4) — each story's acceptance criteria must reference named audit fields from §8
- synthetic-phi-generator (Execution Stage C, Skill 5) — threshold constants in §3 drive test data generation
- ECC (Claude Code) — pseudocode in §4 is the implementation blueprint
- harness-audit-grader (Execution Stage D, Skill 6) — §8 audit schema is the SD-01B contract

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Feature Logic Summary

When a carer calls in sick, the Home-Care-AI vacancy incident pipeline automatically finds the best replacement, presents a ranked shortlist to the care coordinator for one-tap approval, and — once approved — dispatches all notifications in the legally required order. The pipeline is event-driven: a `vacancy_recorded` event triggers the entire flow. It begins with a hard qualification gate (no expired credentials, no scheduling conflicts), then scores qualified carers on two weighted dimensions: proximity to the client (40%) and SPP match quality (60%, sourced from the client's Soft Preference Profile — visit history, familiarity threshold, binary acceptance flag). The coordinator is the human-in-the-loop (HITL): the system never assigns a carer without coordinator approval. If the coordinator does not respond within 30 minutes, a backup coordinator is notified. If neither responds, the incident is escalated to `VACANCY_UNRESOLVED` — the system never auto-assigns and never leaves an incident in a hanging state. After approval, the system sends the assignment SMS to the carer (CC-8 compliant — no PHI), the briefing SMS after confirmation (CC-6 compliant — no match explanation), the client notification (G-DS-05 phrasing branch), and finally the family notification — but only after the client has been notified (E-3 Gate 2: the Arthur Kovacs constraint). Every state transition produces an immutable audit log entry. All external content is template-interpolated — no LLM generates text sent to carers, clients, or families.

Inputs / States / Outputs

Inputs

```
// Trigger
Input: vacancy_data.client_id // Type: UUID | Source: coordinator-recorded or carer-reported
Input: vacancy_data.visit_time // Type: Timestamp ISO 8601 UTC | Source: scheduling system
Input: vacancy_data.absent_carer_id // Type: UUID | Source: absence record
Input: vacancy_data.care_requirements // Type: String[] | Source: client record (structured qualifications list)

// Client SPP (Soft Preference Profile)
Input: client.familiarity_threshold // Type: Enum ['KNOWN_CARERS_ONLY', 'BRIEFED_ACCEPTABLE', 'ANY'] | Source: SPP P-3
Input: client.spp_completeness_score // Type: Float 0.0-1.0 (P-10) | Source: computed field
Input: client.suburb // Type: String (suburb name) | Source: SPP P-11 (suburb only – never full address)
Input: client.notification_channel // Type: String (phone number) | Source: F-1 enrolment
Input: client.family_contact_channel // Type: String (phone number) | Source: F-2 enrolment
Input: client.cognitive_flag // Type: Boolean (P-4) | Source: SPP (consent-gated)
Input: client.personal_sensitivities // Type: String max 100 chars (P-5) | Source: SPP

// Carer Records
Input: carer.qualifications // Type: String[] | Source: HR database
Input: carer.credential_expiry_dates // Type: Map<credential, Date> | Source: HR database (Wave 1 dormant: carer.credential_expiry_date)
Input: carer.availability_calendar // Type: TimeSlot[] | Source: scheduling system
Input: carer.postcode // Type: String | Source: carer self-reported (S-2)
Input: carer.phone_number // Type: String | Source: employment contract (S-1b, tokenized reference)

// Computed SPP history
Input: p7_visit_count(carer_id, client_id) // Type: Integer | Source: visit log
Input: p8_binary_flag(carer_id, client_id) // Type: Boolean | Source: visit outcome log
```

States

```
State: NORMAL // No active vacancy incident
State: VACANCY_DETECTED // vacancy_recorded event received; input validation passed
State: SHORTLIST_READY // Qualification + scoring complete; ranked shortlist generated
State: HITL_PENDING // Coordinator push notification sent; SLA timer running
State: COORDINATOR_APPROVED // Coordinator tapped Approve; coordinator_approved = true
State: CARER_NOTIFIED // Assignment SMS sent; carer reply timer running
State: BRIEFING_SENT // Carer replied YES; briefing SMS sent
State: CLIENT_NOTIFIED // Client notification SMS delivered; client_notified = true
State: FAMILY_NOTIFIED // Family notification SMS delivered (E-3 Gate 2 passed)
State: RESOLVED // Incident closed; resolution screen shown to coordinator
State: VACANCY_UNRESOLVED // Terminal failure state – no candidate assigned; coordinator + agency owner alerted
```

Outputs

```
Output: push_notification(recipient, message, urgency_level) // Level 1 – informational
Output: serve_shortlist(coordinator_id, shortlist, vacancy_data) // Level 1 – display only
Output: serve_approval_card(coordinator_id, candidate, preview) // Level 1 – display only
Output: send_sms(channel, recipient, template, vars, content_generation_method) // Level 3 – external action
Output: log_event(APPAuditLogEntry) // Level 1 – immutable write
Output: alert_coordinator(message, urgency_level) // Level 1 – internal push
Output: escalate_vacancy_unresolved(case_id, incident_data) // Level 3 – Sonnet; agency owner notified
Output: alert_pm_lead(message) // Level 1 – rolling threshold breach
```

Threshold Constants

All constants named here must appear verbatim in the implementation. Any change to a constant value requires a PR comment with justification. Named constants are referenced in every audit log entry's `threshold_applied` field.

```
// Scoring weights (must sum to 1.0)
CONST SCORING_WEIGHT_FAMILIARITY = 0.60 // P-7/P-8 familiarity history drives 60% of score
CONST SCORING_WEIGHT_PROXIMITY = 0.40 // Geographic proximity drives 40% of score
// Rationale: Angela (CC-001) interview: "Who the client knows matters more than how far they travel."
// Review cadence: after E1 validation of OKR-3 (≥70% 1-tap trust rate)

// Shortlist
CONST SHORTLIST_MAX_CANDIDATES = 3 // Maximum candidates shown to coordinator
CONST PROXIMITY_MAX_KM = 25 // Maximum carer distance (km) to be included in shortlist
// Rationale: Sydney metro + outer suburbs. Carers >25km away unlikely to arrive on time.
// Review cadence: after geographic coverage data from first 20 agencies

// SLA timers
CONST COORDINATOR_APPROVAL_SLA_MIN = 30 // Minutes – primary coordinator HITL window
CONST BACKUP_COORDINATOR_SLA_MIN = 15 // Minutes – backup coordinator HITL window
CONST CARER_REPLY_GRACE_MIN = 15 // Minutes – carer YES/NO reply window
// Rationale: Artifact 15 S3 workflow. Angela's "45-minute morning cascade" sets the ceiling.
// Review cadence: after E1 carer reply-rate data (XP-4A)

// SPP cold-start
CONST SPP_COLD_START_THRESHOLD = 0 // P-10 = 0 triggers EC-04 path
// Rationale: Zero SPP completeness → matching falls back to proximity-only. Coordinator card shows invitation.

// Familiarity scoring normalisation
CONST FAMILIARITY_VISIT_CAP = 5 // Visit counts above 5 treated as maximum familiarity (1.0)
CONST FAMILIARITY_BINARY_FLOOR = 0.80 // P-8 acceptance flag floors familiarity score at 0.80

// Compliance feature flags (defaults – toggled only by Privacy Officer or Legal sign-off)
CONST E1_LEGAL_SIGNOFF = false // P-2 excluded from scoring until anti-discrimination opinion obtained
CONST WHATSAPP_APP8_CONFIRMED = false // WhatsApp channel locked until APP 8 DPA confirmed
CONST SC07_GOOGLE_MAPS_APPROVED = false // Google Maps API locked until APP 8 review complete (AX-01 action)
// Note: SC07_GOOGLE_MAPS_APPROVED does NOT block shortlist generation – proximity falls back to POSTCODE_ZONE mode

// Audit + compliance
CONST AUDIT_LOG_RETENTION_YEARS = 7 // HIPAA-grade floor – append-only; no DELETE or UPDATE

// VACANCY_UNRESOLVED rolling threshold
CONST VACANCY_UNRESOLVED_ALERT_THRESHOLD = 0.10 // 10% over 14-day rolling window → PM Lead L1 alert
CONST VACANCY_UNRESOLVED_WINDOW_DAYS = 14 // Rolling window for alert threshold calculation
// Rationale: Artifact 22 BS-4. Above 10% indicates match algorithm degradation requiring PM review.

// Schema enforcement
CONST P9_FREE_TEXT_FIELD_PERMITTED = false // Always false – CRIT-04. Any schema with this field is CRITICAL defect.

// Model selection (HS-STRAT-03 binding)
CONST TOKEN_BUDGET_L1_L2 = 2200 // Haiku: all routine orchestration
CONST TOKEN_BUDGET_L3 = 3500 // Sonnet: VACANCY_UNRESOLVED escalation only
CONST MODEL_L1_L2 = 'claude-haiku-4-5-20251001'
CONST MODEL_L3 = 'claude-sonnet-4-6'

// F10 dormant fields (Wave 1 schema – display deferred to P1.1, data planted Wave 1)
// These are schema field names – not logic constants – but named here for ECC visibility
// DORMANT: carer.credential_expiry_date → powers F10 Compliance Dashboard (P1.1)
// DORMANT: client.care_plan_review_due_date → powers F10 Compliance Dashboard (P1.1)
// DORMANT: visit.documentation_complete → powers F10 Compliance Dashboard (P1.1)
```

Pseudocode Logic Gates

```
// =====
// HOME-CARE-AI – VACANCY INCIDENT PROCESSING PIPELINE
// Model: Haiku (L1/L2) unless explicitly noted as Sonnet (L3)
// Content generation: TEMPLATE_INTERPOLATION only (CRIT-02)
// =====

ON_EVENT vacancy_recorded(vacancy_data):

    // -----
    // GATE 0 – Input Validation + Idempotency Guard
    // -----
    IF vacancy_data.client_id IS NULL OR vacancy_data.visit_time IS NULL OR
       vacancy_data.absent_carer_id IS NULL:
        log_event(event_type='INPUT_VALIDATION_FAILED',
                  state_before='NORMAL', state_after='NORMAL',
                  action_taken='DISCARD_EVENT',
                  ai_confidence_score=1.0, model_id=MODEL_L1_L2)
        alert_coordinator("Incomplete vacancy data received – please re-submit the absence.")
        RETURN

    IF duplicate_vacancy_exists(vacancy_data.client_id, vacancy_data.visit_time):
        log_event(event_type='DUPLICATE_VACANCY_SUPPRESSED',
                  state_before='NORMAL', state_after='NORMAL',
                  action_taken='DISCARD_DUPLICATE')
        RETURN // Idempotency guard – prevents double-processing

    SET case_id = generate_uuid_v4()
    SET incident_state = 'VACANCY_DETECTED'
    log_event(event_type='VACANCY_DETECTED',
              state_before='NORMAL', state_after='VACANCY_DETECTED',
              case_id=case_id, action_taken='INCIDENT_OPENED')

    // -----
    // GATE 1 – Qualification Filter (ACT-V-02 – hard binary, pass/fail)
    // -----
    qualified_carers = []

    FOR EACH carer IN agency_roster:
        IF carer.id == vacancy_data.absent_carer_id:
            SKIP // Do not re-assign the absent carer

        IF NOT carer.available_at(vacancy_data.visit_time):
            SKIP // Scheduling conflict

        IF has_expired_credential(carer, vacancy_data.care_requirements):
            SKIP // Hard block – no expired credentials regardless of match quality
            // Note: carer.credential_expiry_date is a Wave 1 schema field (F10 dormant)

        IF NOT qualifications_match(carer, vacancy_data.care_requirements):
            SKIP // Required qualification not held

        qualified_carers.append(carer)

    IF qualified_carers.length == 0:
        // EC-01: No qualified carers in roster for this visit
        GOTO Gate 14 // VACANCY_UNRESOLVED – no candidates to offer coordinator

    // -----
    // GATE 2 – SPP Cold-Start Check (EC-04)
    // P-10 = 0 → all clients in this incident have zero SPP completeness
    // -----
    client = fetch_client_record(vacancy_data.client_id)

    IF client.spp_completeness_score == SPP_COLD_START_THRESHOLD: // P-10 = 0
        SET briefing_mode = 'MINIMAL'
        log_event(event_type='SPP_COLD_START',
                  state_before='VACANCY_DETECTED', state_after='VACANCY_DETECTED',
                  action_taken='BRIEFING_MODE_SET_MINIMAL',
                  guard_id='EC-04', guard_passed=true)
        // Note: SPP cold-start does NOT block matching. Scoring falls back to proximity-only.
        // Coordinator card must show: "Add [client.first_name]'s preferences now – 2 min" (DR-4 compliant)
        // This is EC-04 – distinct from EC-02 (roster exhaustion) and EC-03 (partial SPP)
    ELSE:
        SET briefing_mode = 'FULL'

    // -----
```

```

// GATE 3 – Proximity Scoring (ACT-V-03)
// APP 8 note: Google Maps transmits carer.postcode to US servers (SC-07 pending)
// Fallback to POSTCODE_ZONE mode until SC07_GOOGLE_MAPS_APPROVED = true
// -----
FOR EACH carer IN qualified_carers:
  IF SC07_GOOGLE_MAPS_APPROVED == true:
    // Only carer.postcode transmitted to Google Maps API – never client address, never client name
    distance_km = google_maps_distance_matrix(carer.postcode, client.suburb)
    ASSERT google_maps_payload_contains(['carer_postcode']) ONLY // CC-8 pre-check on API call
  ELSE:
    // Postcode zone fallback – no external API call required, no APP 8 trigger
    distance_km = postcode_zone_distance(carer.postcode, client.suburb)
    // Note: postcode_zone_distance() uses a pre-loaded AU postcode→suburb zone mapping
    // Accuracy: ±3km for metro areas. Acceptable for v1 shortlist ranking.

  IF distance_km > PROXIMITY_MAX_KM:
    qualified_carers.remove(carer) // Outside acceptable radius
    CONTINUE

  // Normalise: closer = higher score (1.0 = same suburb, 0.0 = PROXIMITY_MAX_KM)
  carer.proximity_score = max(0.0, 1.0 - (distance_km / PROXIMITY_MAX_KM))

IF qualified_carers.length == 0:
  // EC-01 variant: all qualified carers are outside proximity radius
  GOTO Gate 14 // VACANCY_UNRESOLVED

// -----
// GATE 4 – SPP Match Scoring (ACT-V-04)
// Weights: familiarity 0.60, proximity 0.40
// G-CC-4: P-2 excluded until E1_LEGAL_SIGNOFF = true
// -----
carers_to_remove = []

FOR EACH carer IN qualified_carers:
  familiarity_score = 0.0

  IF briefing_mode == 'FULL':
    // P-7: Visit count with this specific client
    visit_count = p7_visit_count(carer.id, client.id)
    IF visit_count > 0:
      familiarity_score = min(1.0, visit_count / FAMILIARITY_VISIT_CAP)

    // P-8: Binary acceptance flag (client has explicitly accepted this carer before)
    IF p8_binary_flag(carer.id, client.id) == true:
      familiarity_score = max(familiarity_score, FAMILIARITY_BINARY_FLOOR) // Floor at 0.80

    // P-3: Familiarity threshold hard filter
    // "Known carers only" clients: unfamiliar carers are disqualified entirely
    IF client.familiarity_threshold == 'KNOWN_CARERS_ONLY' AND visit_count == 0:
      carers_to_remove.append(carer) // Hard filter – cannot be overridden by proximity
      CONTINUE

  // G-CC-4 guard: P-2 advisory only, never scored
  // Named constant default: E1_LEGAL_SIGNOFF = false
  IF E1_LEGAL_SIGNOFF == false:
    p2_weight = 0.0 // P-2 excluded from scoring_weights
    // P-2 is displayed on coordinator card as: "Client preference (advisory – not scored)"
    log_event(event_type='COMPLIANCE_OVERRIDE_ACKNOWLEDGED',
              guard_id='G-CC-4', guard_passed=true,
              action_taken='P2_EXCLUDED_FROM_SCORING')
    // If E1_LEGAL_SIGNOFF later becomes true, p2_weight assignment changes here –
    // matching engine rebuild required; CRIT-03 constraint formally lifted

  carer.composite_score = (familiarity_score * SCORING_WEIGHT_FAMILIARITY) +
    (carer.proximity_score * SCORING_WEIGHT_PROXIMITY)

  // Apply P-3 hard filter
FOR EACH carer IN carers_to_remove:
  qualified_carers.remove(carer)

IF qualified_carers.length == 0:
  // P-3 'KNOWN_CARERS_ONLY' filter eliminated all candidates
  // This is the KNOWN_CARERS_ONLY + no-familiar-carers-available edge case
  // Coordinator must manually find someone the client knows, or manage client expectations
  GOTO Gate 14 // VACANCY_UNRESOLVED

// -----
// GATE 5 – Shortlist Generation (ACT-V-06)
// Maximum SHORTLIST_MAX_CANDIDATES candidates, ranked by composite_score descending
// -----
shortlist = qualified_carers

```

```

        .sort(by='composite_score', order='descending')
        .take(SHORTLIST_MAX_CANDIDATES)

// CC-1 compound guard: P-3 + P-4 + P-5 must never appear together in any external payload
// Coordinator shortlist payload contains: carer name, familiarity count, qualification badge, proximity
// It does NOT contain P-4 (cognitive flags) or P-5 (personal sensitivities) – those are in briefing only
// Assert pre-build that the shortlist payload assembly function enforces this
ASSERT coordinator_shortlist_payload_builder.contains_p4 == false // guard_id='CC-1'
ASSERT coordinator_shortlist_payload_builder.contains_p5 == false // guard_id='CC-1'

// RAG architectural pre-specification (HIGH-05)
// If any future feature retrieves SPP fields from a vector store:
// MANDATORY: filter = {'client_id': vacancy_data.client_id}
// Any retrieval query without this filter MUST raise a runtime error (not silently proceed)
// This is NFR-05 – specified here even though no RAG feature is in v1

log_event(event_type='SHORTLIST_GENERATED',
          state_before='VACANCY_DETECTED', state_after='SHORTLIST_READY',
          action_taken='SHORTLIST_BUILT',
          ai_confidence_score=shortlist[0].composite_score)
SET incident_state = 'SHORTLIST_READY'

// -----
// GATE 6 – Coordinator HITL Notification (ACT-V-05)
// Level 1 action – push notification only; no action taken without coordinator tap
// -----
log_event(event_type='HITL_REQUESTED',
          state_before='SHORTLIST_READY', state_after='HITL_PENDING',
          action_taken='COORDINATOR_NOTIFIED',
          hitl_id=primary_coordinator.id, reviewer_role='CARE_COORDINATOR')
SET incident_state = 'HITL_PENDING'

push_notification(
  recipient = primary_coordinator,
  message = "[absent_carer.first_name] called in sick – [N] visits affected. Replacements ready for review.",
  urgency_level = 'HIGH'
)

// Session isolation requirement (HIGH-06)
// The coordinator's shortlist session is scoped to this case_id only.
// On session expiry or logout, shortlist data is cleared from coordinator app memory.
// Per-session encryption key provisioned for this case.
SET coordinator_session = create_session(case_id, primary_coordinator.id, expires_on_logout=true)

START_SLA_TIMER(coordinator_id=primary_coordinator.id,
                duration=COORDINATOR_APPROVAL_SLA_MIN,
                on_timeout=GOTO Gate_7c_primary_timeout)

// -----
// GATE 7a – Coordinator Approves Top Candidate (Happy Path)
// -----
ON_COORDINATOR_ACTION(coordinator_id, action='APPROVE', candidate_id):
  CANCEL_SLA_TIMER(coordinator_id=primary_coordinator.id)
  selected_carer = fetch_carer(candidate_id)
  SET coordinator_approved = true

// Validate this action is from the active session (session isolation HIGH-06)
ASSERT coordinator_session.case_id == case_id
ASSERT coordinator_session.user_id == coordinator_id

log_event(event_type='COORDINATOR_APPROVED',
          state_before='HITL_PENDING', state_after='COORDINATOR_APPROVED',
          user_id=coordinator_id, hitl_id=coordinator_id,
          hitl_decision='CONFIRMED',
          hitl_response_ms=elapsed_since_hitl_requested(),
          action_taken='CANDIDATE_APPROVED',
          ai_confidence_score=selected_carer.composite_score,
          threshold_applied='SCORING_WEIGHT_FAMILIARITY=0.60|SCORING_WEIGHT_PROXIMITY=0.40')
SET incident_state = 'COORDINATOR_APPROVED'
GOTO Gate 8

// -----
// GATE 7b – Coordinator Selects Non-Top Candidate (Override)
// -----
ON_COORDINATOR_ACTION(coordinator_id, action='OVERRIDE', candidate_id):
  // Override is valid – coordinator has institutional knowledge the algorithm may lack
  CANCEL_SLA_TIMER(coordinator_id=primary_coordinator.id)
  selected_carer = fetch_carer(candidate_id)
  SET coordinator_approved = true

```

```

log_event(event_type='COORDINATOR_OVERRIDE',
          state_before='HITL_PENDING', state_after='COORDINATOR_APPROVED',
          user_id=coordinator_id, hitl_id=coordinator_id,
          hitl_decision='CONFIRMED',
          hitl_notes='Coordinator selected non-top-ranked candidate',
          hitl_response_ms=elapsed_since_hitl_requested(),
          action_taken='CANDIDATE_OVERRIDE_SELECTED')
SET incident_state = 'COORDINATOR_APPROVED'
GOTO Gate 8

// -----
// GATE 7c – Primary Coordinator Timeout (30-min SLA expired)
// -----
ON_SLA_TIMEOUT(coordinator_id=primary_coordinator.id):
  log_event(event_type='HITL_TIMEOUT',
            state_before='HITL_PENDING', state_after='HITL_PENDING',
            reviewer_role='CARE_COORDINATOR',
            hitl_id=primary_coordinator.id,
            hitl_response_ms=null,
            hitl_decision='TIMEOUT',
            action_taken='ESCALATE_TO_BACKUP_COORDINATOR')

  backup_coordinator = fetch_backup_coordinator(agency_id)

  IF backup_coordinator != null:
    push_notification(
      recipient = backup_coordinator,
      message = "[primary_coordinator.first_name] hasn't responded – [N] visits need a replacement approved.
URGENT.",
      urgency_level = 'CRITICAL'
    )
    log_event(event_type='HITL_TIER_ESCALATED',
              state_before='HITL_PENDING', state_after='HITL_PENDING',
              hitl_id=backup_coordinator.id, reviewer_role='CARE_COORDINATOR',
              action_taken='BACKUP_COORDINATOR_NOTIFIED')

    START_SLA_TIMER(coordinator_id=backup_coordinator.id,
                    duration=BACKUP_COORDINATOR_SLA_MIN,
                    on_timeout=GOTO Gate_7d_backup_timeout)

    // Gate 7a/7b logic applies identically for backup_coordinator response
    ON_COORDINATOR_ACTION(coordinator_id=backup_coordinator.id, ...):
      // Proceed as 7a or 7b above
      GOTO Gate 8

  ELSE:
    // No backup coordinator configured – treat as double timeout immediately
    GOTO Gate_7d_backup_timeout

// -----
// GATE 7d – Backup Coordinator Timeout → VACANCY_UNRESOLVED
// -----
ON_SLA_TIMEOUT(coordinator_id=backup_coordinator.id):
  log_event(event_type='HITL_DOUBLE_TIMEOUT',
            state_before='HITL_PENDING', state_after='VACANCY_UNRESOLVED',
            hitl_decision='TIMEOUT',
            hitl_response_ms=null,
            action_taken='ESCALATE_VACANCY_UNRESOLVED')
  GOTO Gate 14 // VACANCY_UNRESOLVED – never auto-assign

// -----
// GATE 8 – Carer Assignment SMS (ACT-C-01)
// Level 3 action – requires coordinator_approved = true (Gate invariant)
// CC-8 guard: no PHI in carer notification payload
// CRIT-02: template interpolation only – no LLM
// -----
ASSERT coordinator_approved == true // Hard gate – never bypassed

// CC-8 guard: build payload from whitelisted fields only
carer_notification_payload = {
  carer_first_name: selected_carer.first_name, // S-1a (first name only)
  agency_name: agency.name,
  client_suburb: client.suburb, // P-11 suburb only – NEVER full address
  visit_time: vacancy_data.visit_time
}

// Explicitly assert no sensitive fields in payload
ASSERT 'client_full_name' NOT IN carer_notification_payload // guard_id='CC-8'
ASSERT 'client_full_address' NOT IN carer_notification_payload // guard_id='CC-8'
ASSERT 'client_id_uuid' NOT IN carer_notification_payload // guard_id='CC-8'

```

```

ASSERT 'spp_fields' NOT IN carer_notification_payload // guard_id='CC-8'
ASSERT 'health_information' NOT IN carer_notification_payload // guard_id='CC-8'
log_event(event_type='CC8_FIELD_STRIPPED',
          guard_id='CC-8', guard_passed=true,
          action_taken='PAYLOAD_SANITISED_PRE_SEND')

// Channel routing (CRIT-01: SMS AU-hosted only until WHATSAPP_APP8_CONFIRMED = true)
ASSERT WHATSAPP_APP8_CONFIRMED == false // Default – locked by Privacy Officer
channel = resolve_channel(WHATSAPP_APP8_CONFIRMED) // Returns 'SMS_AU' if false
// SMS_AU = MessageMedia or AWS SNS ap-southeast-2 – no cross-border transmission

send_sms(
  channel = channel,
  recipient = selected_carer.phone_number,
  template = "Hi {carer_first_name}, {agency_name} needs you to cover a visit for a client in "
            "{client_suburb} at {visit_time}. Reply YES to confirm or NO if unavailable.",
  vars = carer_notification_payload,
  content_generation_method = 'TEMPLATE_INTERPOLATION'
)
ASSERT content_generation_method == 'TEMPLATE_INTERPOLATION' // CRIT-02 – no LLM call

log_event(event_type='CARER_NOTIFIED',
          state_before='COORDINATOR_APPROVED', state_after='CARER_NOTIFIED',
          action_taken='ACT_C_01_SENT',
          cross_border_disclosure=false, app8_basis='Domestic',
          lawful_basis='Operations', data_sensitivity='PII')
SET incident_state = 'CARER_NOTIFIED'

START_SLA_TIMER(carer_id=selected_carer.id,
                duration=CARER_REPLY_GRACE_MIN,
                on_timeout=GOTO Gate_10c_carer_timeout)

// -----
// GATE 9 – Carer Briefing (ACT-C-02)
// RUNS AFTER CARER CONFIRMS (Gate 10a) – not immediately after assignment SMS
// CC-6 guard: no match explanation, no gender preference in briefing payload
// G-DS-05 guard: familiarity phrasing branches on client.familiarity_threshold
// CC-1 guard: P-3+P-4+P-5 never together in any single payload
// -----
FUNCTION assemble_and_send_briefing(client, selected_carer, briefing_mode):

  briefing_payload = {}

  // CC-6 guard: assert no match explanation or gender preference before payload assembly
  ASSERT 'match_explanation' NOT IN briefing_payload // guard_id='G-CC-6'
  ASSERT 'gender_preference' NOT IN briefing_payload // guard_id='G-CC-6'
  ASSERT 'scoring_weights' NOT IN briefing_payload // guard_id='G-CC-6'

  IF briefing_mode == 'FULL':
    // P-5: Personal sensitivities (structured, ≤100 chars – not free text)
    IF client.personal_sensitivities NOT NULL:
      briefing_payload.sensitivities_note = client.personal_sensitivities

    // P-6: Entry protocol
    IF client.entry_protocol NOT NULL:
      briefing_payload.entry_note = client.entry_protocol

    // G-DS-05 phrasing branch: client.familiarity_threshold drives wording
    IF client.familiarity_threshold == 'KNOWN_CARERS_ONLY':
      briefing_payload.familiarity_note = "This client prefers familiar carers – "
                                         "please introduce yourself clearly and take a calm, "
                                         "steady approach."
    ELSE IF client.familiarity_threshold == 'BRIEFED_ACCEPTABLE':
      briefing_payload.familiarity_note = "This client has been briefed that you may be visiting. "
                                         "They know your name."
    // 'ANY' threshold: no special familiarity note required
  ELSE:
    // briefing_mode == 'MINIMAL' (EC-04 cold-start)
    briefing_payload.familiarity_note = "Client preferences have not yet been set up. "
                                         "Please introduce yourself warmly at arrival."

  // CC-1 compound guard: P-3 + P-4 + P-5 must never appear together in external payload
  p3_in_payload = briefing_payload.familiarity_note NOT NULL AND client.familiarity_threshold NOT NULL
  p4_in_payload = client.cognitive_flag == true AND briefing_payload.cognitive_guidance NOT NULL
  p5_in_payload = briefing_payload.sensitivities_note NOT NULL

  IF p3_in_payload AND p4_in_payload AND p5_in_payload:
    // CC-1 violation: full vulnerability profile in a single carer payload
    log_event(event_type='CC6_GUARD_BLOCKED', // Closest canonical event for guard block
              guard_id='CC-1', guard_passed=false,
              action_taken='BRIEFING_SUPPRESSED_CC1_VIOLATION')
    alert_coordinator("Briefing blocked – compound sensitivity guard CC-1 triggered. ")

```

```

        "Review and send manually to avoid disclosing full vulnerability profile.")
    RETURN // Do not send – coordinator must manually manage

    // P-4 (cognitive flag) must be stripped from external payload entirely
    ASSERT 'cognitive_flag' NOT IN briefing_payload // P-4 – NEVER in external payload
    ASSERT 'p4_raw_value' NOT IN briefing_payload // P-4

    log_event(event_type='CC6_GUARD_BLOCKED' IF cc6_fails ELSE 'BRIEFING_SENT',
              guard_id='G-CC-6', guard_passed=true)

    send_sms(
      channel = channel,
      recipient = selected_carer.phone_number,
      template = briefing_template,
      vars = briefing_payload,
      content_generation_method = 'TEMPLATE_INTERPOLATION'
    )
    ASSERT content_generation_method == 'TEMPLATE_INTERPOLATION' // CRIT-02

    log_event(event_type='BRIEFING_SENT',
              state_before='CARER_NOTIFIED', state_after='BRIEFING_SENT',
              action_taken='ACT_C_02_SENT',
              consent_record_id=client.spp_consent_record_id, // HIGH-02: lawful basis for P-5/P-6 disclosure
              lawful_basis='Treatment', data_sensitivity='SENSITIVE_INFO')
    SET incident_state = 'BRIEFING_SENT'

    // -----
    // GATE 10a – Carer Confirms (YES)
    // -----
    ON_CARER_REPLY(carer_id=selected_carer.id, reply='YES'):
      CANCEL_SLA_TIMER(carer_id=selected_carer.id)
      // State transition: CARER_NOTIFIED → BRIEFING_SENT (via assemble_and_send_briefing)
      assemble_and_send_briefing(client, selected_carer, briefing_mode)
      GOTO Gate 11

    // -----
    // GATE 10b – Carer Declines (NO)
    // -----
    ON_CARER_REPLY(carer_id=selected_carer.id, reply='NO'):
      CANCEL_SLA_TIMER(carer_id=selected_carer.id)
      log_event(event_type='CARER_DECLINED',
                state_before='CARER_NOTIFIED', state_after='HITL_PENDING',
                action_taken='SHORTLIST_RE_PRESENTED',
                hitl_decision=null)
      SET incident_state = 'HITL_PENDING'

      // Re-surface shortlist – coordinator must re-approve; no auto-assign
      remaining_shortlist = shortlist.remove(selected_carer)
      SET coordinator_approved = false

      IF remaining_shortlist.length > 0:
        alert_coordinator(
          "[selected_carer.first_name] is unavailable – next best match: "
          "[remaining_shortlist[0].first_name]. Tap to assign.",
          urgency_level = 'HIGH'
        )
        // Coordinator re-approves: return to Gate 7a with remaining_shortlist
        GOTO Gate 7 (remaining_shortlist, same case_id, new SLA timer)

      ELSE:
        // EC-02: All shortlist candidates exhausted
        GOTO Gate 14 // VACANCY_UNRESOLVED

    // -----
    // GATE 10c – Carer No-Reply (SLA timeout)
    // Treated as implicit decline – treated identically to Gate 10b
    // -----
    ON_SLA_TIMEOUT(carer_id=selected_carer.id):
      alert_coordinator(
        "[selected_carer.first_name] hasn't replied in {CARER_REPLY_GRACE_MIN} minutes. "
        "Confirm this carer or pick another candidate.",
        urgency_level = 'HIGH'
      )
      log_event(event_type='CARER_DECLINED',
                state_before='CARER_NOTIFIED', state_after='HITL_PENDING',
                action_taken='CARER_TIMEOUT_TREATED_AS_DECLINE')
      GOTO Gate 10b logic (re-surface shortlist)

    // -----

```

```

// GATE 11 – E-3 Gate 1: Client Notification (ACT-P-01)
// Level 3 action – requires coordinator_approved = true
// G-DS-05: phrasing branches on familiarity threshold + visit history
// -----
ASSERT coordinator_approved == true // Gate invariant
SET client_notified = false

visit_count_for_assigned = p7_visit_count(selected_carer.id, client.id)

// G-DS-05 phrasing branch – see Artifact 16 CC-2 and G-DS-05 guard
IF visit_count_for_assigned > 0:
    client_sms_template = "Good morning, this is {agency_name}. Your visit today will be with "
                        "{carer_first_name}, who has visited you before. "
                        "They'll arrive at {visit_time}."

ELSE IF client.familiarity_threshold == 'BRIEFED_ACCEPTABLE' AND visit_count_for_assigned == 0:
    client_sms_template = "Good morning, this is {agency_name}. Your visit today will be covered "
                        "by {carer_first_name}. They know about your preferences and will "
                        "introduce themselves when they arrive at {visit_time}."

ELSE IF client.familiarity_threshold == 'KNOWN_CARERS_ONLY' AND visit_count_for_assigned == 0:
    // This state MUST NOT be reached – Gate 4 P-3 filter should have prevented this assignment.
    // If somehow reached: it is a Gate 4 logic error – escalate as G-DS-05 violation.
    log_event(event_type='COMPLIANCE_OVERRIDE_ACKNOWLEDGED',
              guard_id='G-DS-05', guard_passed=false,
              action_taken='G_DS05_VIOLATION_DETECTED')
    alert_coordinator("Error: [client.first_name] requires a known carer, but [carer.first_name] "
                     "has no prior visits. Please select a known carer manually.")
    GOTO Gate 14 // VACANCY_UNRESOLVED – constraint cannot be satisfied

// E-3 Gate 1: send client notification
sms_result = send_sms(
    channel = 'SMS_AU',
    recipient = client.notification_channel,
    template = client_sms_template,
    vars = { agency_name, carer_first_name, visit_time },
    content_generation_method = 'TEMPLATE_INTERPOLATION'
)
ASSERT content_generation_method == 'TEMPLATE_INTERPOLATION' // CRIT-02

IF sms_result.delivered == true:
    SET client_notified = true
    log_event(event_type='CLIENT_NOTIFIED',
              state_before='BRIEFING_SENT', state_after='CLIENT_NOTIFIED',
              action_taken='ACT_P_01_SENT',
              lawful_basis='Treatment', data_sensitivity='PII')
    SET incident_state = 'CLIENT_NOTIFIED'
    GOTO Gate 12

ELSE:
    SET client_notified = false
    log_event(event_type='CLIENT_NOTIFICATION_UNAVAILABLE',
              state_before='BRIEFING_SENT', state_after='BRIEFING_SENT',
              action_taken='ACT_P_01_FAILED_COORDINATOR_ALERTED')
    alert_coordinator(
        "[client.first_name] has no notification channel – please call them directly "
        "before family is notified. Tap 'Client notified' once done to release family notification.",
        urgency_level = 'HIGH'
    )
    // Family notification HOLDS until coordinator manually acknowledges client notification
    // client_notified remains false – Gate 12 will block until true

// -----
// GATE 12 – E-3 Gate 2: Family Notification (ACT-F-01)
// Arthur Kovacs constraint – non-bypassable structural gate
// Function signature enforces both preconditions
// -----
FUNCTION send_family_notification(coordinator_approved: bool, client_notified: bool):

    // E-3 Gate 2 – hard structural gate (HIGH-08, HS-STRAT-02f)
    IF NOT (coordinator_approved AND client_notified):
        log_event(event_type='E3_GATE_BLOCKED',
                  state_before='CLIENT_NOTIFIED', state_after='FAMILY_GATE_BLOCKED',
                  guard_id='G-E3-1', guard_passed=false,
                  action_taken='FAMILY_NOTIFICATION_SUPPRESSED')
        alert_coordinator(
            "Family notification held – [client.first_name] has not yet been notified. "
            "Notify them manually and tap 'Client notified' to release.",
            urgency_level = 'MEDIUM'
        )
    RETURN // Do not send – Arthur Kovacs constraint

```

```

// Phrasing variant: familiarity-aware
IF visit_count_for_assigned > 0:
    family_carer_phrase = "who has visited {client_pronoun} before"
ELSE:
    family_carer_phrase = "who knows about {client_first_name}'s care preferences"

family_sms_template = "Hi {family_first_name}, {agency_name} is letting you know that "
                        "{client_first_name}'s visit today will be covered by {carer_first_name}, "
                        "{family_carer_phrase}. Arranged by {coordinator_first_name}. – {agency_name}"

sms_result = send_sms(
    channel = 'SMS_AU',
    recipient = client.family_contact_channel,
    template = family_sms_template,
    vars = { family_first_name, agency_name, client_first_name, carer_first_name,
             family_carer_phrase, coordinator_first_name },
    content_generation_method = 'TEMPLATE_INTERPOLATION'
)
ASSERT content_generation_method == 'TEMPLATE_INTERPOLATION' // CRIT-02

log_event(event_type='FAMILY_NOTIFIED',
          state_before='CLIENT_NOTIFIED', state_after='FAMILY_NOTIFIED',
          guard_id='G-E3-1', guard_passed=true,
          action_taken='ACT_F_01_SENT',
          consent_record_id=client.family_notification_consent_id, // MED-03: consent scope
          lawful_basis='Operations', data_sensitivity='PII',
          cross_border_disclosure=false, app8_basis='Domestic')
SET incident_state = 'FAMILY_NOTIFIED'
GOTO Gate 13

// Call site (E-3 Gate 2 enforced by function signature)
send_family_notification(
    coordinator_approved = coordinator_approved, // bool
    client_notified = client_notified // bool
)

// -----
// GATE 13 – Incident Resolution
// -----
log_event(event_type='INCIDENT_RESOLVED',
          state_before='FAMILY_NOTIFIED', state_after='RESOLVED',
          action_taken='INCIDENT_CLOSED')
SET incident_state = 'RESOLVED'

push_notification(
    recipient = primary_coordinator,
    message = "[selected_carer.first_name] confirmed for [client.first_name] at [visit_time]. "
              "[client.first_name] notified ✓. [family_contact.first_name] notified ✓. "
              "This incident has been logged with timestamp and decision rationale.",
    urgency_level = 'LOW'
)
// Resolution screen serves coordinator per DR-3 (client-centred language, emotional payoff)

// Post-incident: SPP completeness prompt (ACT-S-03)
IF client.spp_completeness_score < 0.80:
    alert_coordinator(
        "Add [client.first_name]'s preferences now – 2 min. It will help next time.",
        urgency_level = 'LOW'
    )

RETURN

// -----
// GATE 14 – VACANCY_UNRESOLVED Escalation
// Level 3 – ONLY place Claude Sonnet 4.6 is called in this pipeline
// -----
FUNCTION escalate_vacancy_unresolved(case_id, incident_data):

    // Assert model constraints (HS-STRAT-03)
    ASSERT model_id == MODEL_L3 // 'claude-sonnet-4-6'
    ASSERT token_budget <= TOKEN_BUDGET_L3 // ≤ 3,500 tokens

    // Build escalation summary using Sonnet
    // System prompt: template-based – no PHI in system prompt itself
    // Input to Sonnet: anonymised incident data (carer UUIDs, NOT names; client UUID only)
    escalation_context = {
        client_id: vacancy_data.client_id, // UUID only – no name
        visit_time: vacancy_data.visit_time,
        candidates_tried_count: shortlist.length,
        decline_reasons: [carer_reply_log], // Reply codes (NO/TIMEOUT) only
        familiarity_threshold: client.familiarity_threshold, // Enum value – not PHI
    }

```

```

    spp_completeness: client.spp_completeness_score // Float – not PHI
}

escalation_summary = sonnet_generate_escalation_summary(
    context = escalation_context,
    system_prompt = VACANCY_UNRESOLVED_SYSTEM_PROMPT // Static template – no PHI
)

// PHI scan before any display or transmission (CRIT-02 – prompt injection defence)
escalation_summary = scan_and_strip_phi(escalation_summary)

log_event(event_type='VACANCY_UNRESOLVED',
          state_before='HITL_PENDING', state_after='VACANCY_UNRESOLVED',
          action_taken='ESCALATE_AGENCY_OWNER',
          model_id=MODEL_L3,
          ai_confidence_score=0.0) // No confidence score applicable
SET incident_state = 'VACANCY_UNRESOLVED'

// Alert coordinator: options presented (no auto-cancel)
push_notification(
    recipient = primary_coordinator,
    message = "No suitable replacement found for [client.first_name] at [visit_time]. "
              "Candidates tried: [N]. Options: Extend search criteria / Call manually / Mark as cancelled.",
    urgency_level = 'CRITICAL'
)

// Alert agency owner
push_notification(
    recipient = agency_owner,
    message = "UNRESOLVED VACANCY: [client.first_name] at [visit_time]. "
              "Manual intervention required. Coordinator has been notified.",
    urgency_level = 'CRITICAL'
)

// Rolling threshold check: alert PM Lead if rate ≥ 10% over 14 days
unresolved_rate = calculate_rolling_unresolved_rate(
    agency_id = agency.id,
    window_days = VACANCY_UNRESOLVED_WINDOW_DAYS // 14
)
IF unresolved_rate >= VACANCY_UNRESOLVED_ALERT_THRESHOLD: // 0.10
    alert_pm_lead(
        "VACANCY_UNRESOLVED rate at [{unresolved_rate * 100:.1f}%] over 14 days at "
        "[agency.name] – above {VACANCY_UNRESOLVED_ALERT_THRESHOLD * 100}% threshold. "
        "Match algorithm review required."
    )
    log_event(event_type='COMPLIANCE_OVERRIDE_ACKNOWLEDGED',
              action_taken='PM_LEAD_THRESHOLD_ALERT_SENT',
              threshold_applied='VACANCY_UNRESOLVED_ALERT_THRESHOLD=0.10')

// System NEVER auto-cancels. Decision authority belongs to coordinator / agency owner.
RETURN

```

SMS Template Registry

Amendment — Implementation Stub Resolution (IS-01)

All v1 notification templates are defined here verbatim. These strings are the **canonical source** for ECC and user stories. Any change to a template requires PM Lead approval + an amendment to this section. Templates are rendered using safe string interpolation only — no LLM, no format strings that accept unvalidated input. Variable names in {braces} map directly to the payload fields defined in §4 Gate 8–12.

Template 1 — ACT-C-01: Carer Assignment SMS

GATE 8 | **Recipient:** Selected carer | **Guard:** CC-8 (no PHI) | **CRIT-02:** Template interpolation only

Hi {carer_first_name}, {agency_name} needs you to cover a visit for a client in {client_suburb} at {visit_time}. Reply YES to confirm or NO if unavailable.

Variable	Source	Type	Compliance Note
{carer_first_name}	selected_carer.first_name (S-1a)	String	First name only — no surname
{agency_name}	agency.name	String	Not a PII field
{client_suburb}	client.suburb (P-11)	String	Suburb only — never full address per CC-8
{visit_time}	vacancy_data.visit_time	String (formatted: "8:00 AM")	Not a PII field

Fields explicitly absent: client full name, client full address, client UUID, any SPP field, any health information.

Template 2 — ACT-C-02: Carer Briefing SMS (3 variants)

GATE 9 (runs after carer replies YES) | **Recipient:** Confirmed carer | **Guards:** CC-6 (no match explanation), G-DS-05 (P-3 phrasing), CC-1 (P-3+P-4+P-5 compound) | **CRIT-02:** Template interpolation only

Variant A — Full SPP, P-3 = BRIEFED_ACCEPTABLE or ANY (briefing_mode = FULL)

Hi {carer_first_name}, here are the details for your visit with {client_first_name} at {visit_time}, {client_full_address}.

{entry_note}
{sensitivities_note}
{familiarity_note}

If you have any questions before the visit, call {coordinator_first_name} at {coordinator_phone}.

Variant B — Full SPP, P-3 = KNOWN_CARERS_ONLY (familiarity_note override)

Same as Variant A with {familiarity_note} set to: "This client prefers familiar carers — please introduce yourself clearly and take a calm, steady approach."

Variant C — Minimal SPP (briefing_mode = MINIMAL — EC-04 cold-start)

Hi {carer_first_name}, you are covering a visit for {client_first_name} at {visit_time}, {client_full_address}.

Client preferences have not yet been set up. Please introduce yourself warmly at arrival.

If you have any questions before the visit, call {coordinator_first_name} at {coordinator_phone}.

Variable	Source	Type	Compliance Note
{carer_first_name}	selected_carer.first_name	String	First name only
{client_first_name}	client.first_name	String	First name only — no surname in v1 briefing
{visit_time}	vacancy_data.visit_time	String	Not PII
{client_full_address}	client.full_address	String	Full address released only post coordinator_approved + carer_confirmed (never in ACT-C-01)
{entry_note}	client.entry_protocol (P-6)	String	Operational field — "Knock and wait" etc. Omitted if null.
{sensitivities_note}	client.personal_sensitivities (P-5)	String chars ≤100	Structured field — not free text. Omitted if null.
{familiarity_note}	Computed from client.familiarity_threshold (P-3)	String	G-DS-05 phrasing branch. Omitted if P-3 = 'ANY' and briefing_mode = 'FULL'.
{coordinator_first_name}	primary_coordinator.first_name	String	Not client PII
{coordinator_phone}	primary_coordinator.phone	String	Agency operational data

Fields explicitly absent from this template

- X Match score or composite_score value
- X Scoring weights or algorithm explanation
- X Gender preference (P-2) — never in any carer communication
- X Cognitive flag (P-4) — never in any external payload
- X Client UUID or any identifier beyond first name
- X Any other SPP field not listed above

Template 3 — ACT-P-01: Client Notification SMS (3 variants, G-DS-05 branch)

GATE 11 | Recipient: Client | Guard: G-DS-05 (phrasing branch), E-3 Gate 1 | CRIT-02: Template interpolation only

Variant A — Carer has prior visits with this client (P-7 visit_count > 0)

Good morning, this is {agency_name}. Your visit today will be with {carer_first_name}, who has visited you before. They'll arrive at {visit_time}.

Variant B — Carer is new to this client, P-3 = BRIEFED_ACCEPTABLE (visit_count = 0)

Good morning, this is {agency_name}. Your visit today will be covered by {carer_first_name}. They know about your preferences and will introduce themselves when they arrive at {visit_time}.

Variant C — P-3 = KNOWN_CARERS_ONLY AND visit_count = 0: UNREACHABLE in normal flow. Gate 4 P-3 hard filter prevents this assignment. If somehow reached → Gate 14 (G-DS-05 violation), not a notification.

Variable	Source	Compliance Note
{agency_name}	agency.name	Not PII
{carer_first_name}	selected_carer.first_name	First name only per CC-8 principle
{visit_time}	vacancy_data.visit_time	Not PII

Fields explicitly absent: carer surname, carer phone, carer address, match score, any SPP field reference, any health information.

Template 4 — ACT-F-01: Family Notification SMS (2 variants, E-3 Gate 2)

GATE

12 | **Recipient:** Client's family contact | **Guard:** G-E3-1 (client_notified = true required) | **CRIT-02:** Template interpolation only

Variant A — Carer has prior visits (P-7 visit_count > 0)

Hi {family_first_name}, {agency_name} is letting you know that {client_first_name}'s visit today will be covered by {carer_first_name}, who has visited {client_pronoun} before. Arranged by {coordinator_first_name}. – {agency_name}

Variant B — Carer is new to this client (visit_count = 0)

Hi {family_first_name}, {agency_name} is letting you know that {client_first_name}'s visit today will be covered by {carer_first_name}, who knows about {client_first_name}'s care preferences. Arranged by {coordinator_first_name}. – {agency_name}

Variable	Source	Compliance Note
{family_first_name}	client.family_contact.first_name (F-2)	First name only
{agency_name}	agency.name	Not PII
{client_first_name}	client.first_name	First name only
{carer_first_name}	selected_carer.first_name	First name only
{client_pronoun}	Computed from client.pronoun_preference (default: "them")	Not sensitive
{coordinator_first_name}	primary_coordinator.first_name	Not PII

Fields explicitly absent: carer surname, carer phone, client full address, any SPP field, any health diagnosis or care needs information, match score.

Template Constants File (B1-A)

SMS templates are **not inline strings** in business logic code. They live in a single versioned constants file:

config/sms_templates.py (or sms_templates.json for language-agnostic access)

Structure:

```
SMS_TEMPLATES = {
    "version": "1.0.0", # Increment on every legal-approved change
    "approved_by": "Privacy Counsel – [Name]",
    "approved_date": "2026-MM-DD", # Copy Approval gate date
    "ACT_C_01": "Hi {carer_first_name}, {agency_name} needs you to cover a visit for a client in {client_suburb} at {visit_time}. Reply YES to confirm or NO if unavailable.",
    "ACT_C_02_VARIANT_A": "Hi {carer_first_name}, here are the details for your visit...", # Full template per §4.1
    "ACT_C_02_VARIANT_B": "...", # KNOWN_CARERS_ONLY variant
    "ACT_C_02_VARIANT_C": "...", # EC-04 minimal variant
    "ACT_P_01_VARIANT_A": "Good morning, this is {agency_name}...", # Prior visits
    "ACT_P_01_VARIANT_B": "Good morning, this is {agency_name}...", # New carer
    "ACT_F_01_VARIANT_A": "Hi {family_first_name}, {agency_name} is letting you know...", # Prior visits
    "ACT_F_01_VARIANT_B": "Hi {family_first_name}, {agency_name} is letting you know...", # New carer
}
```

- version is logged in APPAuditLogEntry.hitl_notes on every send: "sms_template_version=1.0.0" — creates an immutable record of which template string was active at the time of each notification
- Legal review targets this file alone — no code change required to update copy after counsel review
- Any PR that adds a template string directly to business logic is a failing audit finding

Template Character Limits

Template	Max chars	SMS segments	Rationale
ACT-C-01	160	1	Single-segment SMS — carer reply-rate highest for short messages
ACT-C-02	480	3 max	Multi-field assembly; truncate {sensitivities_note} at 100 chars (P-5 limit)
ACT-P-01	160	1	Client may be elderly; short message reduces confusion
ACT-F-01	200	2 max	Family context warrants slightly more detail

If any assembled template exceeds its character limit, the `send_sms()` function must log a LOW severity warning and truncate at the segment boundary — never silently drop fields mid-sentence.

Mermaid Sequence Diagrams

Diagram 1 — Happy Path (Top Candidate Approved, Carer Confirms, All Notifications)

SEQUENCE DIAGRAM

Diagram 1 — Happy Path (Top Candidate Approved, Carer Confirms, All Notifications)

```
sequenceDiagram
    participant CC as Care Coordinator (HITL)
    participant AGENT as AI Agent (Haiku L1/L2)
    participant MATCH as Match Engine
    participant SMS as SMS Gateway AU
    participant AUD as Audit Log (Immutable)

    Note over AGENT: vacancy_recorded event received
    AGENT->>AUD: VACANCY_DETECTED (NORMAL → VACANCY_DETECTED)
    AGENT->>MATCH: Gate 1 – Qualification filter (ACT-V-02)
    MATCH->>AGENT: Qualified carer list (hard credential + availability gate)
    AGENT->>MATCH: Gate 2 – SPP cold-start check (EC-04 branch if P-10 = 0)
    AGENT->>MATCH: Gate 3 – Proximity scoring (postcode zone fallback until SC-07 approved)
    AGENT->>MATCH: Gate 4 – SPP match scoring (P-7/P-8 familiarity; P-2 excluded, G-CC-4)
    MATCH->>AGENT: Ranked shortlist [David Kim 0.87, Sarah Ng 0.61, Marco Torres 0.44]
    AGENT->>AUD: SHORTLIST_GENERATED (VACANCY_DETECTED → SHORTLIST_READY)
    AGENT->>AUD: HITL_REQUESTED (SHORTLIST_READY → HITL_PENDING)
    AGENT->>CC: Push notification: "Jenny sick – 3 visits affected. Replacements ready."
    Note over AGENT,CC: 30-min SLA timer started
    CC->>AGENT: Opens app → reviews David Kim approval card
    Note over CC: Sees: "★ 2 prior visits with Mrs. Kim" + qualifications badge + notification preview (DR-1/DR-2)
    CC->>AGENT: action=APPROVE, candidate=David Kim
    AGENT->>AUD: COORDINATOR_APPROVED (HITL_PENDING → COORDINATOR_APPROVED)
    AGENT->>AGENT: Gate 8 – CC-8 guard: strip sensitive fields from payload
    AGENT->>AUD: CC8_FIELD_STRIPPED (guard_id=CC-8, guard_passed=true)
    AGENT->>SMS: ACT-C-01: "Hi David, CareBridge needs you to cover a visit in Surry Hills at 8AM. YES/NO?"
    AGENT->>AUD: CARER_NOTIFIED (COORDINATOR_APPROVED → CARER_NOTIFIED)
    Note over AGENT,SMS: 15-min carer reply timer
    SMS->>AGENT: David replies YES
    AGENT->>AGENT: Gate 9 – CC-6 guard: no match explanation in briefing
    AGENT->>AUD: CC6_GUARD_APPLIED (guard_id=G-CC-6, guard_passed=true)
    AGENT->>AGENT: G-DS-05 phrasing branch: P-3 + visit count evaluated
    AGENT->>SMS: ACT-C-02: Briefing (entry protocol + P-5 sensitivities + G-DS-05 phrasing)
    AGENT->>AUD: BRIEFING_SENT (CARER_NOTIFIED → BRIEFING_SENT)
    AGENT->>AGENT: Gate 11 – E-3 Gate 1: assert coordinator_approved=true
    AGENT->>SMS: ACT-P-01: "Good morning, your visit today will be with David, who has visited you before."
    AGENT->>AUD: CLIENT_NOTIFIED (BRIEFING_SENT → CLIENT_NOTIFIED, client_notified=true)
    AGENT->>AGENT: Gate 12 – E-3 Gate 2: assert client_notified=true (G-E3-1)
    AGENT->>AUD: E3_GATE_PASSED (guard_id=G-E3-1, guard_passed=true)
    AGENT->>SMS: ACT-F-01: "Hi Margaret, David Kim will cover your mother's visit today..."
    AGENT->>AUD: FAMILY_NOTIFIED (CLIENT_NOTIFIED → FAMILY_NOTIFIED)
    AGENT->>AUD: INCIDENT_RESOLVED (FAMILY_NOTIFIED → RESOLVED)
    AGENT->>CC: Resolution screen: "David confirmed ✓ Mrs. Kim notified ✓ Margaret notified ✓"
```

Diagram 2 — Override + Carer Decline Path (Coordinator Picks #3, Carer Declines, Next Candidate Confirmed)

SEQUENCE DIAGRAM

Diagram 2 — Override + Carer Decline Path (Coordinator Picks #3, Carer Declines, Next Candidate Confirmed)

```
sequenceDiagram
    participant CC as Care Coordinator (HITL)
    participant AGENT as AI Agent (Haiku L1/L2)
    participant SMS as SMS Gateway AU
    participant AUD as Audit Log (Immutable)

    Note over CC: Coordinator opens shortlist – institutional knowledge overrides algorithm
    CC->>AGENT: action=OVERRIDE, candidate=Marco Torres (ranked #3)
    AGENT->>AUD: COORDINATOR_OVERRIDE (HITL_PENDING → COORDINATOR_APPROVED)
```

```

AGENT->>AGENT: Gate 8 – CC-8 guard applied to Marco Torres payload
AGENT->>SMS: ACT-C-01: "Hi Marco, CareBridge needs you to cover a visit in Surry Hills at 8AM..."
AGENT->>AUD: CARER_NOTIFIED (COORDINATOR_APPROVED → CARER_NOTIFIED)
SMS->>AGENT: Marco replies NO (unavailable)
AGENT->>AUD: CARER_DECLINED (CARER_NOTIFIED → HITL_PENDING)
AGENT->>CC: "Marco is unavailable – next best match: David Kim (0.87). Tap to assign."
Note over AGENT,CC: Shortlist re-presented. No auto-assign. Coordinator must re-approve.
Note over AGENT,CC: New 30-min SLA timer for remaining shortlist
CC->>AGENT: action=APPROVE, candidate=David Kim
AGENT->>AUD: COORDINATOR_APPROVED (HITL_PENDING → COORDINATOR_APPROVED)
AGENT->>SMS: ACT-C-01 to David Kim
AGENT->>AUD: CARER_NOTIFIED (COORDINATOR_APPROVED → CARER_NOTIFIED)
SMS->>AGENT: David replies YES
AGENT->>SMS: ACT-C-02 Briefing to David Kim (CC-6 guard, G-DS-05 phrasing)
AGENT->>AUD: BRIEFING_SENT (CARER_NOTIFIED → BRIEFING_SENT)
AGENT->>SMS: ACT-P-01 to Mrs. Kim (E-3 Gate 1)
AGENT->>AUD: CLIENT_NOTIFIED (BRIEFING_SENT → CLIENT_NOTIFIED)
AGENT->>SMS: ACT-F-01 to Margaret (E-3 Gate 2 – client_notified=true confirmed)
AGENT->>AUD: FAMILY_NOTIFIED (CLIENT_NOTIFIED → FAMILY_NOTIFIED)
AGENT->>AUD: INCIDENT_RESOLVED (FAMILY_NOTIFIED → RESOLVED)
AGENT->>CC: Resolution screen

```

Diagram 3 — HITL Timeout → Backup Coordinator → VACANCY_UNRESOLVED

SEQUENCE DIAGRAM

Diagram 3 — HITL Timeout → Backup Coordinator → VACANCY_UNRESOLVED

```

sequenceDiagram
    participant CC as Primary Coordinator
    participant BCC as Backup Coordinator
    participant AGENT as AI Agent (Haiku L1/L2)
    participant SONNET as Claude Sonnet 4.6 (L3 only)
    participant OWNER as Agency Owner
    participant AUD as Audit Log (Immutable)

    AGENT->>AUD: HITL_REQUESTED (SHORTLIST_READY → HITL_PENDING)
    AGENT->>CC: Push notification (HIGH urgency): "Jenny sick – replacements ready for review."
    Note over AGENT,CC: 30-min SLA timer started
    Note over CC: Primary coordinator unavailable (phone off / out of range)
    AGENT->>AUD: HITL_TIMEOUT (HITL_PENDING → HITL_PENDING, hitl_response_ms=null)
    AGENT->>AUD: HITL_TIER_ESCALATED (HITL_PENDING → HITL_PENDING, reviewer=BACKUP_COORDINATOR)
    AGENT->>BCC: Push notification (CRITICAL urgency): "Primary coordinator unresponsive – URGENT: replacements need approval."
    Note over AGENT,BCC: 15-min SLA timer started
    Note over BCC: Backup coordinator also unavailable
    AGENT->>AUD: HITL_DOUBLE_TIMEOUT (HITL_PENDING → VACANCY_UNRESOLVED, hitl_response_ms=null)
    Note over AGENT,SONNET: ONLY Sonnet call in the entire pipeline – L3 escalation
    AGENT->>SONNET: Generate escalation summary (anonymised context; token_budget ≤ 3,500)
    SONNET->>AGENT: Escalation summary text
    AGENT->>AGENT: PHI scan applied to Sonnet output before any display
    AGENT->>AUD: VACANCY_UNRESOLVED (HITL_PENDING → VACANCY_UNRESOLVED, model_id=claude-sonnet-4-6)
    AGENT->>CC: CRITICAL push: "No replacement found for [client] at [visit_time]. Options: extend / call manually / mark as cancelled."
    AGENT->>OWNER: CRITICAL push: "UNRESOLVED VACANCY: [client] at [visit_time]. Manual intervention required."
    Note over AGENT: System never auto-cancels. Incident remains VACANCY_UNRESOLVED until coordinator acts.

    alt Coordinator manually resolves
        CC->>AGENT: Manual candidate selection + approval
        AGENT->>AUD: COORDINATOR_APPROVED (VACANCY_UNRESOLVED → COORDINATOR_APPROVED)
        Note over AGENT: Continue from Gate 8 (carer SMS + notifications)
    else Coordinator cancels visit
        CC->>AGENT: Mark as cancelled (explicit coordinator decision)
        AGENT->>AUD: ASSIGNMENT_CANCELLED (VACANCY_UNRESOLVED → RESOLVED)
    end
end

```

Agentic Autonomy Classification

Every output action classified per CLAUDE.md Article VIII Agentic Control Matrix.

Action	Level	Name	HITL Required	Reversible?	False-Positive Cost	Guard
log_event()	1	Informer	No	N/A — append only	None	—
push_notification(coordinator, ...)	1	Informer	No	Yes (can be dismissed)	Alarm fatigue	—
push_notification(agency_owner, ...)	1	Informer	No	Yes (can be dismissed)	Minor distraction	—
serve_shortlist()	1	Informer	No	Yes (read-only display)	None	—
serve_approval_card()	1	Informer	No	Yes (read-only display)	None	DR-1, DR-2
alert_coordinator()	1	Informer	No	Yes (notification)	Noise	—
alert_pm_lead()	1	Informer	No	Yes (notification)	Minor distraction	BS-4 threshold
send_sms(carer, assignment)	3	Escalator	Yes — coordinator_approved = true	No — SMS sent	Carer unnecessarily mobilised	CC-8, CRIT-01
send_sms(carer, briefing)	3	Escalator	Yes — carer reply YES	No — SMS sent	SPP data disclosed to carer unnecessarily	CC-6, G-DS-05
send_sms(client, notification)	3	Escalator	Yes — coordinator_approved = true	No — SMS sent	Client anxiety from incorrect info	G-DS-05, E-3 Gate 1
send_sms(family, notification)	3	Escalator	Yes — client_notified = true	No — SMS sent	Family notified before client (Arthur Kovacs failure)	G-E3-1, CRIT-02
escalate_vacancy_unresolved()	3	Escalator	Yes — both HITL timeouts exhausted	No — agency owner alerted	Coordinator disrupted for false alarm	Sonnet L3

Level 2 actions are not used in this pipeline. The coordinator approval card (Gate 7) is the single HITL gate that elevates L3 actions. The coordinator IS the HITL — no RN/MD chain required for a scheduling product.

Sonnet usage restriction (HS-STRAT-03 binding): The `escalate_vacancy_unresolved()` function is the **ONLY** call to `claude-sonnet-4-6` in the entire pipeline. Any PR that introduces a Sonnet call outside Gate 14 is a scope violation and requires PM Lead sign-off before merge.

Edge Cases & Failure Modes

ID	Failure Mode	Trigger Condition	System Response	Audit Event	Recovery
EC-01	Empty shortlist — no qualified carers	Qualification gate returns 0 candidates; or all qualified carers outside proximity radius	Immediate VACANCY_UNRESOLVED escalation (Gate 14) — coordinator alerted before any SLA timer starts	VACANCY_UNRESOLVED	Coordinator manually expands search criteria (increase PROXIMITY_MAX_KM, relax qualification filter)
EC-02	Roster exhaustion	All SHORTLIST_MAX_CANDIDATES carers decline or timeout sequentially	After last candidate decline: VACANCY_UNRESOLVED escalation (Gate 14). Coordinator sees full decline log.	CARER_DECLINED × N → VACANCY_UNRESOLVED	Coordinator manual outreach outside shortlist; or reschedule visit
EC-03	Partial SPP (some fields empty)	client.spp_completeness_score > 0 AND < 0.80	Matching proceeds with available fields. Briefing omits empty field sections. Coordinator sees completeness indicator on approval card. Post-resolution SPP prompt fires (ACT-S-03).	BRIEFING_SENT (partial payload)	Coordinator populates missing SPP fields via completeness prompt. No matching failure.
EC-04	SPP cold-start — zero SPP	client.spp_completeness_score == SPP_COLD_START_THRESHOLD (P-10 = 0)	SPP_COLD_START event logged. briefing_mode = MINIMAL. Familiarity flag removed from coordinator approval card. Proximity-only scoring used. Approval card shows: "Add [client]'s preferences now — 2 min" (DR-4 invitation, not error).	SPP_COLD_START	Coordinator populates SPP immediately via in-card invitation link
EC-05	Client notification channel unavailable	client.notification_channel IS NULL or SMS delivery fails	CLIENT_NOTIFICATION_UNAVAILABLE logged. Family notification held (E-3 Gate 2 blocked). Coordinator alerted: "Call [client.first_name] directly. Tap 'Client notified' to release family notification."	CLIENT_NOTIFICATION_UNAVAILABLE + E3_GATE_BLOCKED	Coordinator calls client; taps confirmation in app; client_notified = true; Gate 12 re-evaluates
EC-06	Carer SMS delivery failure	send_sms(carer) returns delivered = false	CARER_NOTIFICATION_FAILED logged. L1 alert to coordinator: "SMS to [carer] failed — call them directly by [visit_time - 30 min]." No auto-retry beyond one attempt in v1.	CARER_NOTIFICATION_FAILED	Coordinator calls carer directly. If carer confirms verbally: coordinator taps "Manually confirmed" in app to advance state machine
EC-07	Duplicate vacancy event	Same client_id + visit_time received twice (network retry, double-submission)	DUPLICATE_VACANCY_SUPPRESSED logged. Second event discarded. No second shortlist generated.	DUPLICATE_VACANCY_SUPPRESSED	Idempotency guard prevents duplicate processing. No coordinator action required.

APP-Aligned Audit Log Schema

Full Schema (Source of Truth for SD-01B)

Base: CLAUDE.md Article IX HIPAAAuditLogEntry. Extended: Artifact 16 §7 APPAuditLogEntry. This combined schema is the contract validated by harness-audit-grader SD-01B.

```
interface APPAuditLogEntry extends HIPAAAuditLogEntry {
  // -----
  // INHERITED from CLAUDE.md Article IX HIPAAAuditLogEntry
  // All fields below marked * are non-nullable – missing = CRITICAL finding
  // -----

  log_id: string; // * UUID v4 – unique per entry, immutable
  timestamp: string; // * ISO 8601 UTC with milliseconds e.g. "2026-03-28T06:30:11.847Z"
  session_id: string; // * UUID – groups all events within one coordinator session
  case_id: string; // * UUID – groups all events within one vacancy incident

  patient_id: string; // * UUID – maps to client UUID; never plaintext name or DOB
  user_id: string; // UUID of coordinator or system actor; null if AI-initiated
  hitl_id: string; // UUID of assigned coordinator reviewer; null until escalated
  reviewer_role: string; // 'CARE_COORDINATOR' | 'BACKUP_COORDINATOR' | 'AGENCY_OWNER' | 'SYSTEM'

  event_type: string; // * See event type registry below (canonical 31-event set)
  state_before: string; // * Agent state before transition
  state_after: string; // * Agent state after transition
  action_taken: string; // * Output action executed e.g. 'ACT_C_01_SENT', 'SHORTLIST_BUILT'

  ai_confidence_score: number; // * Float 0.0–1.0 (1.0 for rule-based logic; composite_score for match events)
  model_id: string; // Model identifier e.g. 'claude-haiku-4-5-20251001' or 'claude-sonnet-4-6'
  trigger_sensor: string; // 'VACANCY_EVENT' | 'CARER_SMS_REPLY' | 'COORDINATOR_TAP' | 'SLA_TIMEOUT'
  trigger_value: string | null; // e.g. 'YES' (carer reply) | 'APPROVE' (coordinator action) | null
  threshold_applied: string; // Named constant e.g. 'COORDINATOR_APPROVAL_SLA_MIN=30'

  hitl_decision: string; // 'CONFIRMED' | 'OVERRIDE' | 'TIMEOUT' | null
  hitl_response_ms: number; // ms between HITL_REQUESTED and coordinator response; null if timeout
  hitl_notes: string; // Coordinator free text (override reason); null if system-generated

  lawful_basis: string; // 'Treatment' | 'Operations' | 'Emergency'
  data_sensitivity: string; // 'PHI' | 'PII' | 'SENSITIVE_INFO' | 'OPERATIONAL' | 'SYNTHETIC'
  entry_hash: string; // * SHA-256 of all fields above
  previous_hash: string; // * Hash of prior entry (hash-chain tamper detection)

  // -----
  // EXTENSIONS – APP-specific fields (Artifact 16 §7)
  // -----

  // APP 8 – Cross-border disclosure tracking
  cross_border_disclosure: boolean; // true if data transmitted to overseas recipient
  recipient_jurisdiction: string; // 'AU' | 'US' (flagged) | 'INTERNAL'
  app8_basis: string; // 'Domestic' | 'Consent' | 'Substantially_Similar' | 'DPA_Confirmed' | 'NA'

  // Consent traceability (HIGH-01 – sensitive information fields)
  consent_record_id: string; // UUID of the consent event authorising this data use; null if not applicable
  consent_version: string; // Version of privacy notice at time of consent

  // APP-aligned sensitivity classification
  // 'SENSITIVE_INFO' = Australian Privacy Act sensitive information (health info under s 6 Privacy Act 1988)
  // Overrides CLAUDE.md 'PHI' classification where APP terminology is more precise

  // Compound combination guard trace
  guard_id: string; // e.g. 'G-CC-6' | 'G-E3-1' | 'CC-1' | 'CC-8' | 'G-DS-05' | 'G-CC-4' | null
  guard_passed: boolean; // true = guard check passed; false = guard blocked the action
  guard_block_reason: string; // Populated if guard_passed = false; null otherwise
}
```

Schema Enforcement Rules

- **Non-nullable fields:** `log_id`, `timestamp`, `patient_id`, `event_type`, `state_before`, `state_after`, `action_taken`, `ai_confidence_score`, `entry_hash`, `previous_hash`. A log entry missing any of these fields is a **CRITICAL** compliance finding (SD-01B merge block).
- `entry_hash` **computation:** SHA-256 computed after all other fields are set. Must cover all fields, not a subset. Computed deterministically from a canonical JSON serialisation (keys alphabetically sorted, no whitespace).
- `previous_hash` **of first entry:** SHA-256 of the `session_id` string alone (session anchor).
- **Immutable store:** All log entries write to AWS CloudTrail + write-once S3 (ap-southeast-2). No application role may have DELETE or UPDATE access on any log table or bucket. Any such permission is a CRITICAL security finding.
- `patient_id` = **UUID only**. Never plaintext name, date of birth, or any identifier that can directly re-identify the client. Violation = CRITICAL (PHI in log).
- `guard_id` + `guard_passed` **mandatory** for any action gated by a compound combination guard (CC-1, CC-4, CC-6, CC-8, G-E3-1, G-DS-05). These fields power the harness-audit-grader SD-02 compliance guard coverage check.
- `cross_border_disclosure` = `false` for all v1 events. If any event records `true`, it must also populate `app8_basis` with a value other than 'Domestic' — triggering an automatic compliance review flag.
- `hitl_response_ms` = `null` only for timeout events. Any non-timeout coordinator action must record elapsed milliseconds — this is OKR-3 instrumentation data ($\geq 70\%$ 1-tap approval rate, time-to-fill OKR-1).
- **7-year retention:** `AUDIT_LOG_RETENTION_YEARS` = 7. Lifecycle policy on S3 bucket must match this constant. Any lower retention configuration is a MEDIUM finding.

Canonical Event Type Registry (Artifact 16 §7 — 31 Events)

Source: Artifact 16 §7 extended event registry + CLAUDE.md Article IX base registry adapted for scheduling domain. This spec cannot invent event type names not on this list (Artifact 22 BS-1). Valid extensions for new edge cases must be added to Artifact 16 §7 first.

event_type	state_before	state_after	Pipeline Gate
VACANCY_DETECTED	NORMAL	VACANCY_DETECTED	Gate 0
SHORTLIST_GENERATED	VACANCY_DETECTED	SHORTLIST_READY	Gate 5
HITL_REQUESTED	SHORTLIST_READY	HITL_PENDING	Gate 6
COORDINATOR_APPROVED	HITL_PENDING	COORDINATOR_APPROVED	Gate 7a
COORDINATOR_OVERRIDE	HITL_PENDING	COORDINATOR_APPROVED	Gate 7b
HITL_TIMEOUT	HITL_PENDING	HITL_PENDING	Gate 7c
HITL_TIER_ESCALATED	HITL_PENDING	HITL_PENDING	Gate 7c → 7d
HITL_DOUBLE_TIMEOUT	HITL_PENDING	VACANCY_UNRESOLVED	Gate 7d
CARER_NOTIFIED	COORDINATOR_APPROVED	CARER_NOTIFIED	Gate 8
CARER_DECLINED	CARER_NOTIFIED	HITL_PENDING	Gate 10b/10c
BRIEFING_SENT	CARER_NOTIFIED	BRIEFING_SENT	Gate 9
CLIENT_NOTIFIED	BRIEFING_SENT	CLIENT_NOTIFIED	Gate 11

event_type	state_before	state_after	Pipeline Gate
FAMILY_NOTIFIED	CLIENT_NOTIFIED	FAMILY_NOTIFIED	Gate 12
E3_GATE_BLOCKED	CLIENT_NOTIFIED	FAMILY_GATE_BLOCKED	Gate 12
FAMILY_NOTIFICATION_SUPPRESSED	CLIENT_NOTIFIED	FAMILY_GATE_BLOCKED	Gate 12 (no family channel)
CC6_GUARD_BLOCKED	COORDINATOR_APPROVED	BRIEFING_BLOCKED	Gate 9
CC8_FIELD_STRIPPED	(any)	(any)	Gate 8
CARER_NOTIFICATION_FAILED	COORDINATOR_APPROVED	(any)	Gate 8 delivery failure
BRIEFING_DELIVERY_FAILED	CARER_NOTIFIED	(any)	Gate 9 delivery failure
CLIENT_NOTIFICATION_UNAVAILABLE	BRIEFING_SENT	(any)	Gate 11 delivery failure
VACANCY_UNRESOLVED	HITL_PENDING	VACANCY_UNRESOLVED	Gate 14
INCIDENT_RESOLVED	FAMILY_NOTIFIED	RESOLVED	Gate 13
ASSIGNMENT_CANCELLED	(any)	RESOLVED	Manual cancel by coordinator
DUPLICATE_VACANCY_SUPPRESSED	NORMAL	NORMAL	Gate 0
SPP_COLD_START	VACANCY_DETECTED	VACANCY_DETECTED	Gate 2
SPP_FIELD_UPDATED	(any)	(any)	ACT-S-03 post-incident prompt
SPP_CONSENT_RECORDED	(any)	(any)	Consent event at intake
P9_COLLECTION_BLOCKED	(any)	(any)	Schema validation on any free-text field attempt
COMPLIANCE_OVERRIDE_ACKNOWLEDGED	(any)	(any)	G-CC-4 applied; threshold alert; G-DS-05 violation
FAMILIARITY_THRESHOLD_OVERRIDE	HITL_PENDING	COORDINATOR_APPROVED	Coordinator explicitly overrides P-3 filter
CONSENT_RECORD_CREATED	(any)	(any)	Consent event created for sensitive information collection

Implementation Notes for ECC (Claude Code)

Model Selection — Binding Constraint (HS-STRAT-03)

- `claude-haiku-4-5-20251001` handles all L1/L2 orchestration (every gate except Gate 14). Token budget hard ceiling: 2,200. Assert `token_count <= TOKEN_BUDGET_L1_L2` before every Haiku call.
- `claude-sonnet-4-6` is invoked **only** in Gate 14 (`escalate_vacancy_unresolved()`). Token budget hard ceiling: 3,500. This is the only Sonnet call in the entire pipeline. Any other Sonnet usage requires PM Lead sign-off.
- **No LLM** for any notification content (ACT-C-01, ACT-C-02, ACT-P-01, ACT-F-01), audit log entries, or shortlist display text. All external content uses `TEMPLATE_INTERPOLATION`. CRIT-02 constraint. Violation = CRITICAL finding at `harness-audit-grader`.
- Enable prompt caching Day 1 for the static Haiku system prompt (vacancy processing instructions that don't change per incident). Token cost reduction target per Artifact 20 \$12.

Compliance Guards — Must Be Code Assertions, Not Comments

The following must appear as runtime assertions in the implementation. A comment describing the guard is NOT a guard. `harness-audit-grader` SD-02 checks for assertion presence:

```
# NFR-01 (CRIT-01): Channel routing lock
assert WHATSAPP_APP8_CONFIRMED == False # Default until Privacy Officer toggles
channel = 'SMS_AU' if not WHATSAPP_APP8_CONFIRMED else resolve_whatsapp_channel()

# NFR-03 (CRIT-03): P-2 excluded from scoring
assert E1_LEGAL_SIGNOFF == False # Default until Legal Counsel sign-off
assert 'p2_gender_preference' not in scoring_weights

# NFR-04 (CRIT-04): P-9 schema exclusion
assert 'free_text_notes' not in spp_schema_fields # Must fire at schema validation layer

# NFR-02 (CRIT-02): No LLM for external content
assert action.content_generation_method == 'TEMPLATE_INTERPOLATION' # On every send_sms() call

# HIGH-07 (CC-6 guard): No match explanation in briefing
assert 'match_explanation' not in briefing_payload
assert 'gender_preference' not in briefing_payload

# HIGH-08 (E-3 gate): Family notification preconditions
def send_family_notification(coordinator_approved: bool, client_notified: bool):
    assert coordinator_approved and client_notified # Hard gate – not an if-check

# HS-STRAT-02g (CC-1): Compound P-3+P-4+P-5 not together in external payload
assert not (p3_in_payload and p4_in_payload and p5_in_payload)
```

Infrastructure Requirements

- **AWS Lambda (ap-southeast-2):** Event-driven function per vacancy incident. Step Functions for state machine orchestration. State transitions write to DynamoDB before any downstream action.
- **DynamoDB:** SPP and carer data. Encryption at rest (SSE-KMS, per-agency customer-managed key). `client_id` as partition key for all SPP queries — no cross-client queries permitted.
- **Write-once S3 + CloudTrail:** All `APPAuditLogEntry` writes go here. Object Lock (COMPLIANCE mode, `AUDIT_LOG_RETENTION_YEARS = 7`). No application role with `s3:DeleteObject` or `s3:PutObjectVersionTagging` on this bucket.

- **SMS Gateway:** MessageMedia or AWS SNS ap-southeast-2. Channel Wrapper architecture — `channel_config.v1 = 'SMS_AU'` as configuration parameter. Never hardcode the channel string in business logic.
- **Google Maps API (ACT-V-03):** Gated behind `SC07_G00GLE_MAPS_APPROVED = false`. Postcode zone fallback table deployed Day 1 for v1. Only `carer.postcode` transmitted to Google API when approved — no client identifiers, no client address.
- **PM Lead Alert Channel — `alert_pm_lead()` destination (IS-02):** AWS SNS topic (`HOME_CARE_AI_PM_ALERTS`), subscribed to PM Lead email address. Environment variable: `PM_LEAD_ALERT_SNS_TOPIC_ARN` (injected via Lambda environment — never hardcoded). Alternative channel: Slack webhook via `PM_LEAD_ALERT_WEBHOOK_URL` env variable (optional, evaluated at runtime if set). The `alert_pm_lead()` function tries SNS first; if SNS publish fails, falls back to webhook if configured; logs `COMPLIANCE_OVERRIDE_ACKNOWLEDGED` with `action_taken='PM_LEAD_ALERT_SENT'` on success or `action_taken='PM_LEAD_ALERT_FAILED'` on both failing. A failed PM Lead alert is LOW severity — it does not block the vacancy pipeline and does not retry (next vacancy incident will re-evaluate the rolling rate).
- **Postcode Zone Distance Table — `postcode_zone_distance()` reference data (IS-03):** Static CSV bundled in the Lambda deployment artifact at `data/au_postcode_centroids.csv`. Source: [Australian Postcode Dataset](https://data.gov.au) (open licence, Australian Bureau of Statistics). Schema: `postcode (string), suburb (string), state (string), lat (float), lng (float)`. One row per postcode; ~18,000 rows; file size ~2.8 MB. Distance computation: Haversine formula on lat/lng centroid pairs — returns km to ±1km accuracy for metro areas, ±3km for regional. The file is loaded once on Lambda cold start and cached in `/tmp/au_postcodes.pkl` (serialised dict keyed by postcode string). Update cadence: annually when ABS releases updated postcode boundaries (typically August). Engineering action: re-bundle updated CSV before annual Lambda deployment. No external API call is made in postcode fallback mode — this satisfies APP 8 (no cross-border data transmission for proximity computation).

Data Schema Requirements

- **No P-9 field of any name** in the SPP schema. Schema migration scripts must include a check: `ASSERT 'free_text_notes' NOT IN table.columns AND 'notes' NOT IN table.columns`. Free-text fields trigger AX-01 engineering confirmation re-review.
- **F10 dormant fields** must be planted as null-permitted columns in Wave 1 schema: `carer.credential_expiry_date (Date)`, `client.care_plan_review_due_date (Date)`, `visit.documentation_complete (Boolean, default null)`. They will be populated post-beachhead when F10 Compliance Dashboard (P1.1) is built — but the schema columns must exist from Wave 1 to avoid a disruptive migration later.
- **Session isolation (HIGH-06):** Coordinator session state (shortlist, approval card) scoped to `case_id`. Per-session encryption key from AWS KMS. Explicit session clear on coordinator logout or app background timeout (10 minutes). Never persist shortlist to browser local storage or device cache.
- `patient_id = client UUID throughout`. Never the client's name or any re-identifiable field. Enforce at the ORM/query layer — no raw SQL that joins `patient_id` to `client_name` in the same result set for logging purposes.

Unit Test Requirements (per Gate)

Every gate must have a dedicated unit test:

Test ID	Gate	Test Scenario	Pass Condition
UT-G0a	Gate 0	Null <code>client_id</code> in <code>vacancy_data</code>	Returns early; <code>INPUT_VALIDATION_FAILED</code> logged

Test ID	Gate	Test Scenario	Pass Condition
UT-G0b	Gate 0	Duplicate vacancy event (same client + time)	DUPLICATE_VACANCY_SUPPRESSED logged; second shortlist NOT generated
UT-G1a	Gate 1	All carers have expired credentials	Returns qualified_carers = []; GOTO Gate 14
UT-G1b	Gate 1	Carer with correct qualifications and availability	Returns non-empty qualified_carers
UT-G2a	Gate 2	client.spp_completeness_score = 0 (EC-04)	briefing_mode = 'MINIMAL'; SPP_COLD_START logged
UT-G4a	Gate 4	client.familiarity_threshold = 'KNOWN_CARERS_ONLY' and all carers have visit_count = 0	All carers removed; GOTO Gate 14
UT-G4b	Gate 4	E1_LEGAL_SIGNOFF = false	P-2 weight = 0; COMPLIANCE_OVERRIDE_ACKNOWLEDGED logged with guard_id='G-CC-4'
UT-G6a	Gate 6	Primary coordinator approves within 30 min	COORDINATOR_APPROVED logged; hitl_response_ms populated
UT-G7c	Gate 7c	Primary coordinator timeout (30 min, no response)	HITL_TIMEOUT logged; backup coordinator notified
UT-G7d	Gate 7d	Both coordinators timeout	HITL_DOUBLE_TIMEOUT logged; GOTO Gate 14
UT-G8a	Gate 8	Payload contains client_full_name	CC-8 assertion fires; test expects AssertionError
UT-G9a	Gate 9	match_explanation in briefing payload	CC-6 assertion fires; CC6_GUARD_BLOCKED logged
UT-G11a	Gate 11	P-3 = 'KNOWN_CARERS_ONLY' AND visit_count = 0 for assigned carer	COMPLIANCE_OVERRIDE_ACKNOWLEDGED with G-DS-05; GOTO Gate 14
UT-G12a	Gate 12	client_notified = false	E3_GATE_BLOCKED logged; family SMS not sent
UT-G12b	Gate 12	client_notified = true	Family SMS sent; FAMILY_NOTIFIED logged
UT-G14a	Gate 14	VACANCY_UNRESOLVED triggered	Model = claude-sonnet-4-6; VACANCY_UNRESOLVED logged; agency owner alerted
UT-G14b	Gate 14	Unresolved rate = 0.12 over 14 days	PM Lead alert fires; VACANCY_UNRESOLVED_ALERT_THRESHOLD=0.10 in threshold_applied

Integration Test Requirements (Synthetic PHI Scenarios)

Per Artifact 22 PRIO-06 (BS-3): the synthetic-phi-generator (Artifact 25) must produce exactly these 5 scenarios. The instruction to ECC is: "Build this feature so it passes the tests in sensors.json, including these scenarios."

Scenario ID	Test Name	Setup	Expected Outcome
SYN-01	P-7 familiarity trust match	Client has P-7 visit_count = 3 for carer David; familiarity_threshold = 'BRIEFED_ACCEPTABLE'	David ranked #1 (familiarity $0.60 \times 0.60 = 0.36$ contribution); approval card shows "3 prior visits"; client SMS uses "who has visited you before" phrasing
SYN-02	VACANCY_UNRESOLVED path	All 3 shortlist candidates decline; coordinator SLA also expires	CARER_DECLINED $\times 3 \rightarrow$ HITL_DOUBLE_TIMEOUT $\rightarrow$ VACANCY_UNRESOLVED; Sonnet call with correct token budget; agency owner alerted; no auto-assign
SYN-03	G-DS-05 suppression	Client P-3 = 'KNOWN_CARERS_ONLY'; only unfamiliar carers available	Gate 4 removes all candidates; VACANCY_UNRESOLVED triggered without attempting carer contact; G-DS-05 guard_id in audit log
SYN-04	Data drift (SPP decay)	Client SPP populated (5 fields) at T=0; all field values set to null at T+30 days to simulate coordinator deletion	Matching at T+30 falls back to proximity-only scoring; SPP_COLD_START event logs; EC-04 path executed; familiarity flag absent from coordinator card

Scenario ID	Test Name	Setup	Expected Outcome
SYN-05	Cold-start EC-04	New client with P-10 = 0; no SPP fields; any familiarity_threshold	SPP_COLD_START logged; briefing_mode = MINIMAL; coordinator card shows "Add preferences now" invitation (DR-4); no match explanation references SPP tags; BRIEFING_SENT uses minimal template

SD-01B Merge Gate Checklist

The harness-audit-grader will verify all of the following before issuing PASS. Any missing item is a defect that blocks merge:

- Every state transition in every pipeline gate writes an APPAuditLogEntry
- All 10 non-nullable fields populated in every log entry
- state_before → state_after matches the canonical event type registry (§8 table)
- entry_hash computed from all fields and present in every entry
- previous_hash present and forms a valid chain (first entry anchored to session_id hash)
- Log entries written to write-once S3 store — NOT to a mutable DynamoDB table
- No application role with DELETE or UPDATE permission on the audit bucket
- guard_id + guard_passed present on all guarded actions (CC-1, CC-4, CC-6, CC-8, G-E3-1, G-DS-05)
- E3_GATE_BLOCKED fires before any FAMILY_NOTIFIED event when client_notified = false
- CC6_GUARD_BLOCKED fires before any BRIEFING_SENT event when match explanation is in payload
- CC8_FIELD_STRIPPED fires before every CARER_NOTIFIED event
- HITL_DOUBLE_TIMEOUT present in every VACANCY_UNRESOLVED path trace (never skipped)
- Sonnet model used only for VACANCY_UNRESOLVED events — Haiku for all others
- content_generation_method = 'TEMPLATE_INTERPOLATION' on every CARER_NOTIFIED, BRIEFING_SENT, CLIENT_NOTIFIED, FAMILY_NOTIFIED event

Pre-Sprint 1 Gates (must be satisfied before any code is written):

- **AX-01:** Engineer confirms P-9 free-text field is absent from v1 data model (CRIT-04 resolution)
- **AX-02 result known:** Either bi-directional AlayaCare write-back is scoped (v2 roadmap) or deferred. Either outcome is acceptable — the unknown is the risk. **If write-back unavailable → Backlog stories US-14 (Manual Absence Record) and US-15 (Bulk SPP Import) enter Sprint 2.** (due 2026-04-01)
- **SC-07:** Legal confirms Google Maps API DPA against APP 8 (due 2026-04-01) — note: fallback already specified; this gate unlocks Google Maps, not the pipeline itself
- **Privacy Counsel briefed:** DPIA for CC-1 compound (P-3+P-4+P-5) initiated before any SPP data collection begins (HIGH-03)
- **3-Tap wireframes approved (sketch-gate):** Minimum acceptable fidelity — hand-drawn or lo-fi sketch approved by PM Lead showing: (a) element vertical order on approval card, (b) relative font size annotations for DR-1, (c) tap target sizes ≥ 44×44px for all primary actions, (d) notification preview positioned above the Approve button (DR-2). Full Figma polish not required to begin Wave 4 coding — sketch-gate sufficient. Required before US-10 coding begins.
- **Copy Approval (B1-B):** Privacy Counsel confirms ACT-C-01, ACT-C-02, ACT-P-01, and ACT-F-01 template strings (Artifact 23 §4.1) are APP-compliant. Coordinator proxy Angela (CC-001) has reviewed all four templates for clarity ("Would you feel comfortable sending this to a client?"). Approval recorded in Copy Doc (see §4.1 amendment note). No ACT-C/ACT-P/ACT-F story passes harness-audit-grader SD-01B until this gate is closed.

This artifact is the Source of Truth for all Stage B and Stage C work. If any threshold constant or pseudocode gate changes, user stories (Artifact 24) must be reviewed for impact, and synthetic-phi-generator scenarios (Artifact 25) must be re-validated against the changed constant.